0.5m Lidar Target Stand Instructions

Tools

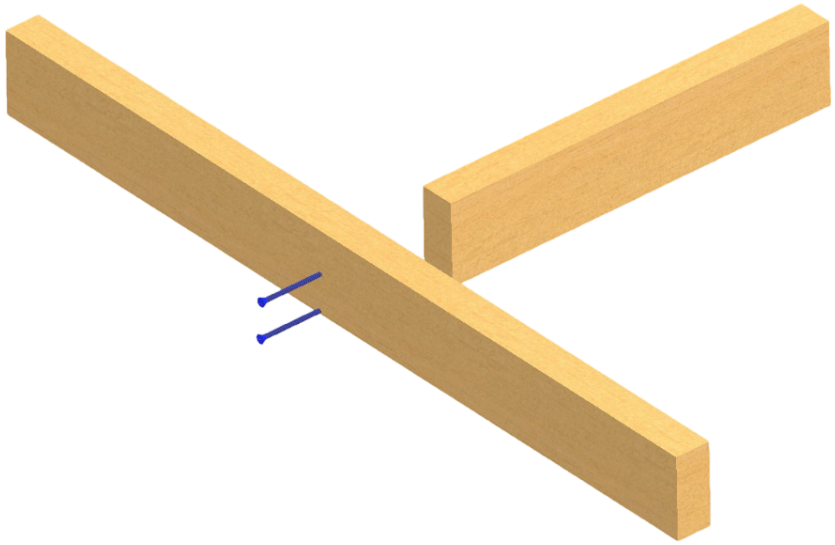
- Cordless Drill
- 5/16-inch Twist Bit
- Chop saw (to cut down lumber)
- Metal saw (to cut down signpost)
- 3D Printer

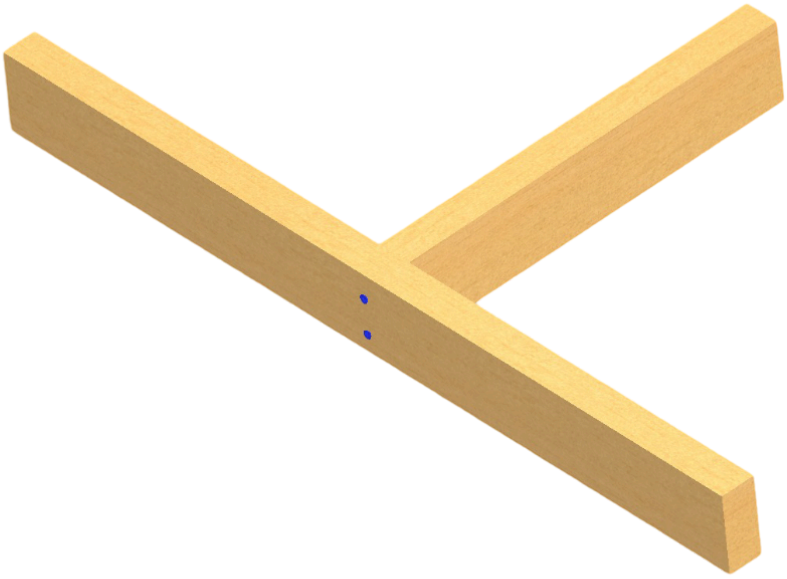
Materials

Item	**Quantity**
2x4 lumber, 17.25-inch length	2
2x4 lumber, 36-inch length	1
2x6 lumber, 8-inch length	1
4-inch x ⅞-inch x 4-inch Corner L-brace	2
2 ½ inch Deck Screws	8
1-inch Wood Screws	4
5/16-inch, 3 ½-inch long Hex Bolts	2
5/16 inch Washer	12
5/16 inch Wing Nut	2
6-foot signpost (7 ft post cut down to 6 ft)	1
3D-printed target bracket for middle support	2
5/16-inch, 2-inch-long Hex Bolt	4
5/16-inch Hex Nut	4

Step 1

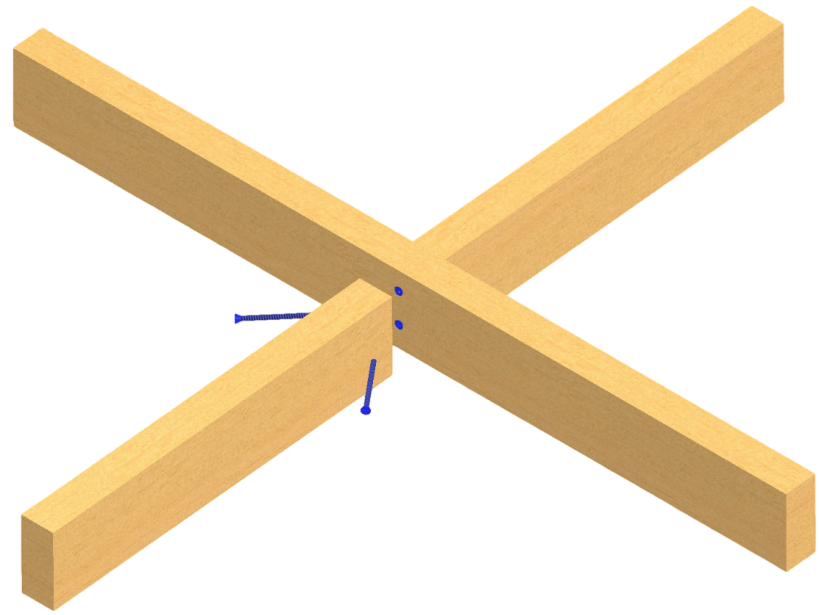
Screw a 17.25-inch 2x4 to the center of the 36-inch 2x4 to form a T-shape, ensuring the smaller board is centered 17.25 inches from each end of the 36-inch board:

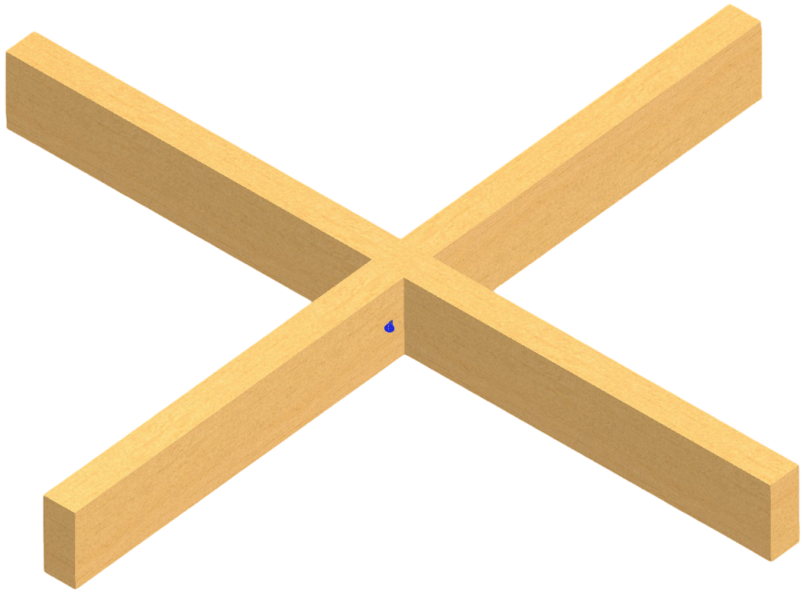




Step 2

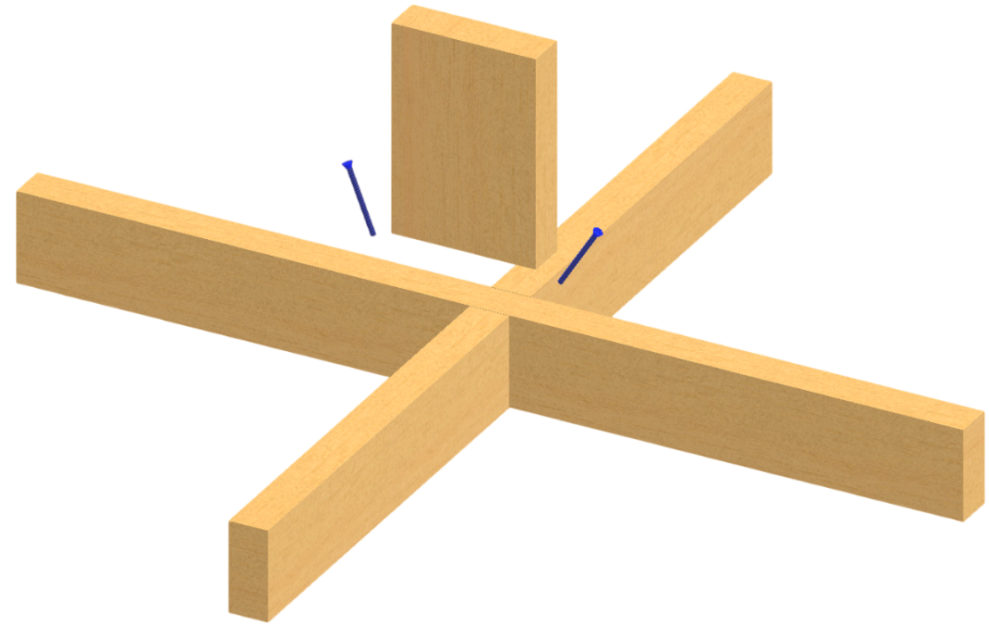
Screw the second 17.25-inch stud to the T-assembly, aligning it with the stud from the previous step to form an X-shape. Drive screws from the side at an angle and use pilot holes if needed:

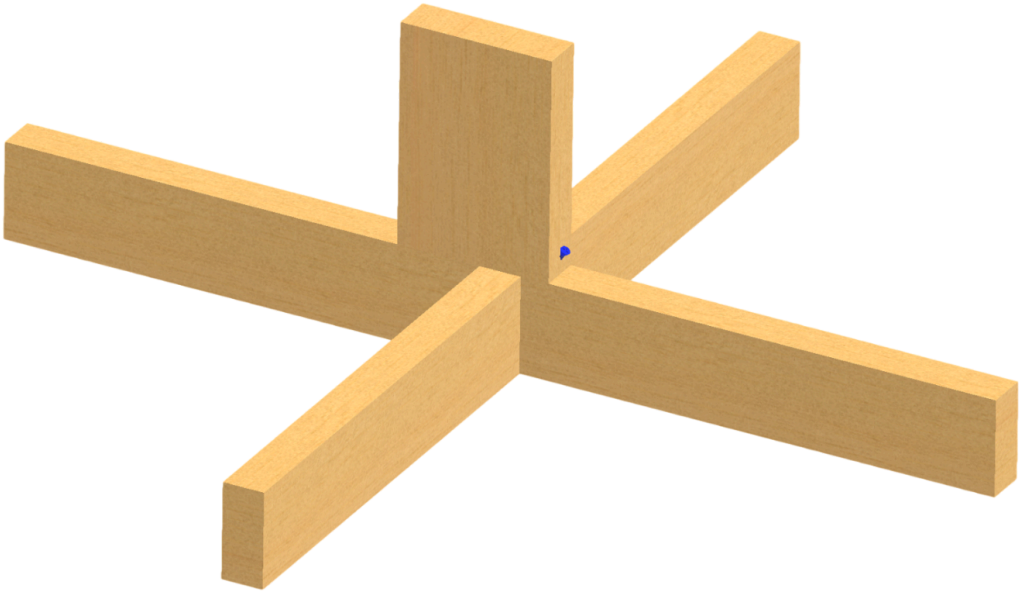




Step 3

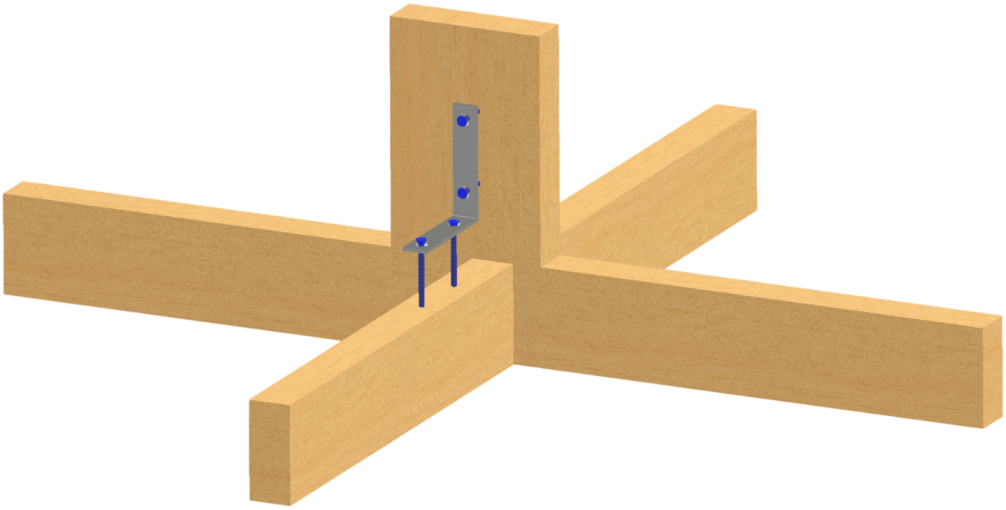
Screw the 2x6 board to the 36-inch board with 2 deck screws. Align the 2x6 so it is 1 inch from the centered cross brace on one side and 3 inches from the cross brace on the other side. Drive the screws from the side at an angle and use pilot holes if needed:

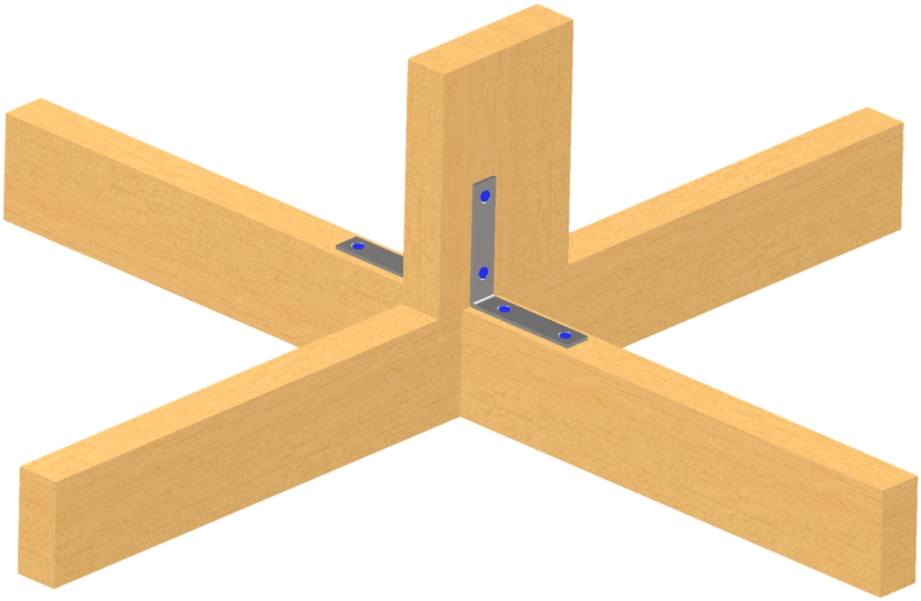




Step 4

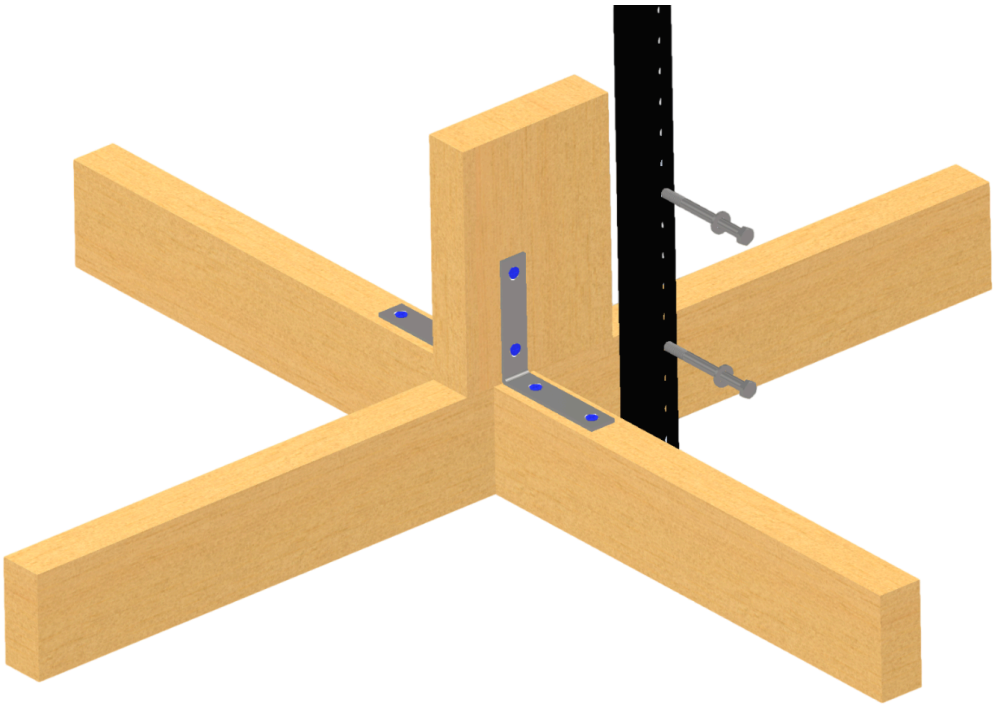
Strengthen the 2x6 connection by screwing a 4-inch metal corner brace to the smaller side boards with the 2 ½-inch deck screws, and then screwing the brace to the 2x6 with the 1-inch wood screws. Repeat for the opposite side:

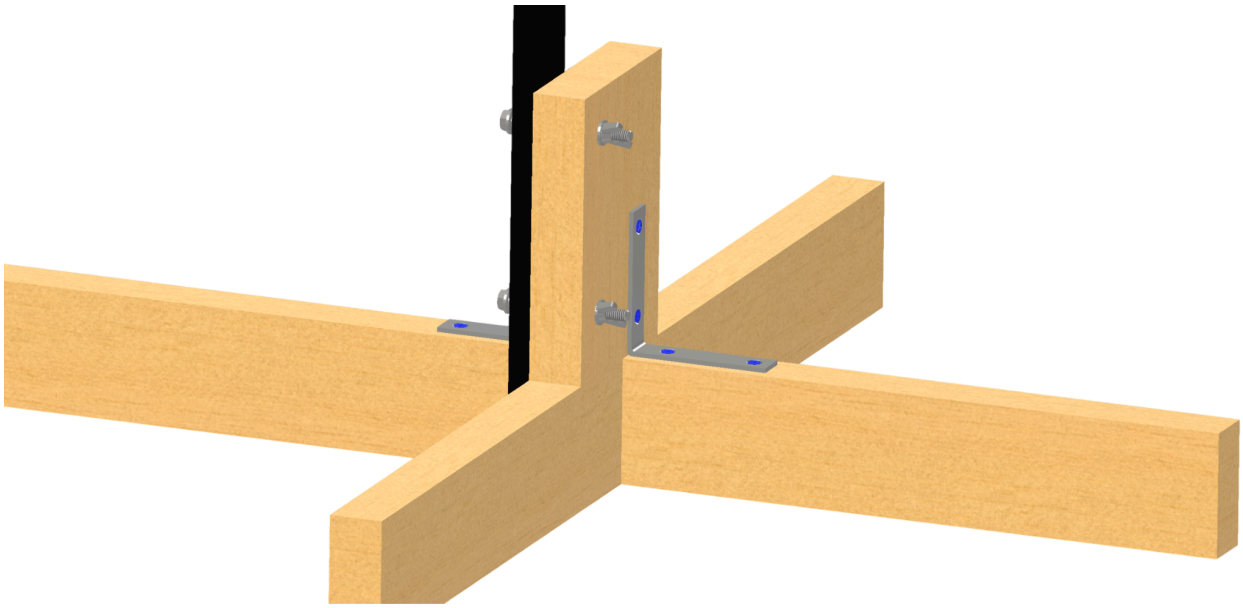




Step 5

Align the signpost against the 2x6 board in the corner of the bottom brace. Drill one hole near the bottom of the 2x6 and another near the top to bolt the post to the base. Then install the 3 ½-inch bolts through the post and wooden base, using a washer on each side, and tighten against the base with a wing nut:





Step 6

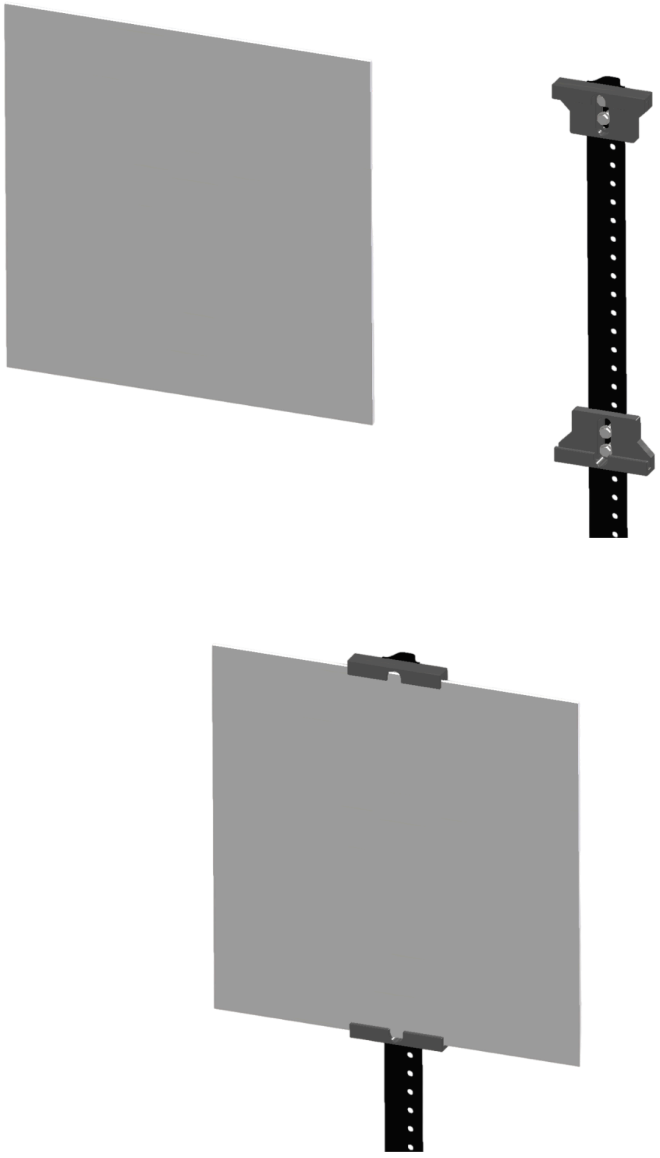
At the top of the signpost, mount the two 3D-printed brackets that will hold the target. Mount the upper bracket as close to the top as possible, and align the lower bracket so that there is a 0.5 m gap between the two trays that will support the target. Mount each bracket with two 2-inch bolts, with a washer on each side of the post, and tighten with a hex nut.





Step 7

Slide the target into the gap left by both 3D-printed brackets, adjusting each bracket as needed to ensure a snug fit:



Completed Assembly

